Time-Delay Mechanisms and Phase-Only Optimization for Phased-Array Parametric Loudspeakers

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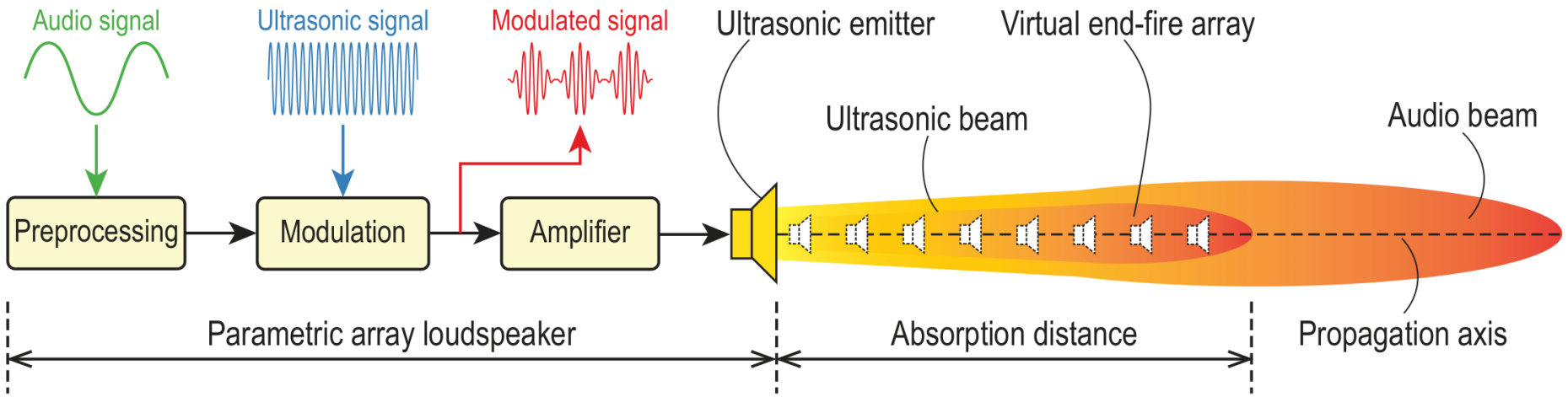
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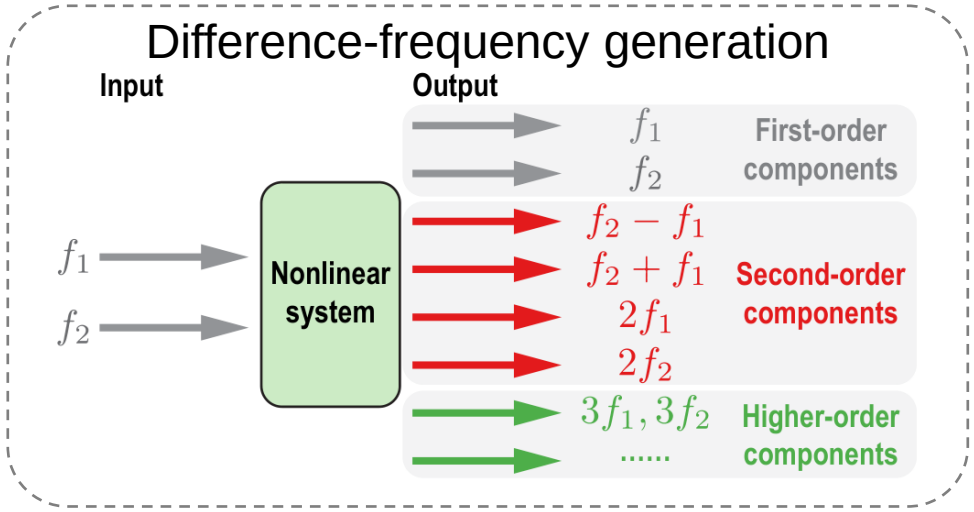
Outline

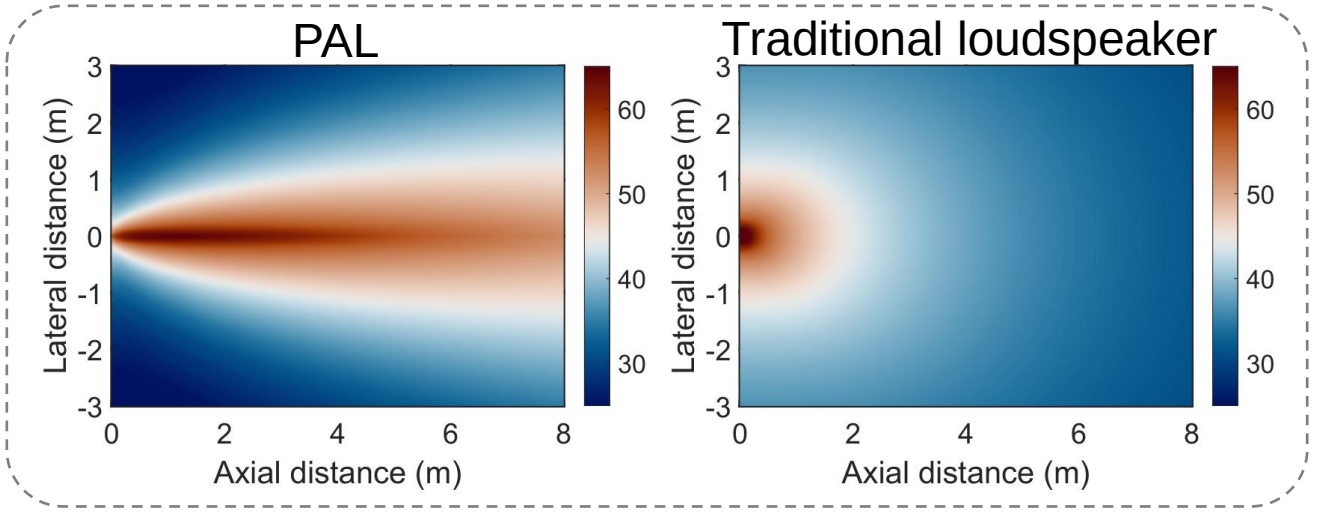
- Introduction
  - Directional audio generated by parametric array loudspeakers (PAL)
  - Quasi-linear modeling of PAL
- Comparative analysis of time-delay strategies
- Phase-only optimization under transmitter constraint
- Conclusions

Directional beams generated by parametric array loudspeakers (PAL)

- Nonlinear demodulation of audio sound from ultrasound
- Highly directional audio beams







[1] J. Zhong and X. Qiu, Acoustic Waves Generated by Parametric Array Loudspeakers, 1st edn. Boca Raton: CRC Press, 2024.
[2] W.-S. Gan, J. Yang, and T. Kamakura, “A review of parametric acoustic array in air,” Applied Acoustics, vol. 73, no. 12, pp. 1211–1219, Dec. 2012.

Quasi-linear modeling of PAL

- Complexity due to the nonlinear process
- Near field: general 2nd nonlinear equation; Kuznetsov equation

$$\left(\nabla^2 - \frac{1}{c_0^2}\frac{\partial^2}{\partial t^2}\right)p + \frac{\delta}{c_0^4}\frac{\partial^3 p}{\partial t^3} = -\frac{\beta}{\rho_0 c_0^4}\frac{\partial^2 p^2}{\partial t^2} - \left(\nabla^2 + \frac{1}{c_0^2}\frac{\partial^2}{\partial t^2}\right)\mathcal{L}$$

Lagrangian density

$$\mathcal{L} = \frac{\rho_0}{2}\mathbf{v}\cdot\mathbf{v} - \frac{p^2}{2\rho_0 c_0^2}$$

- Westervelt far field: Westervelt equation ($\mathcal{L} = 0$)
- Inverse-law far field: simplified models
- Quasilinear approximation: nonlinear equation $\Longrightarrow$ two coupled linear equations ($\mathcal{L} = 0$)

$$\left(\nabla^2 + k_i^2\right)p_i(\mathbf{r}) = 0, \qquad i = 1, 2, \qquad k_i = \frac{\omega_i}{c_0} + \mathrm{i}\alpha(\omega_i) \qquad \left(\nabla^2 + k_\mathrm{a}^2\right)p_\mathrm{a}(\mathbf{r}) = \mathrm{i}\rho_0\omega_\mathrm{a}q(\mathbf{r})$$

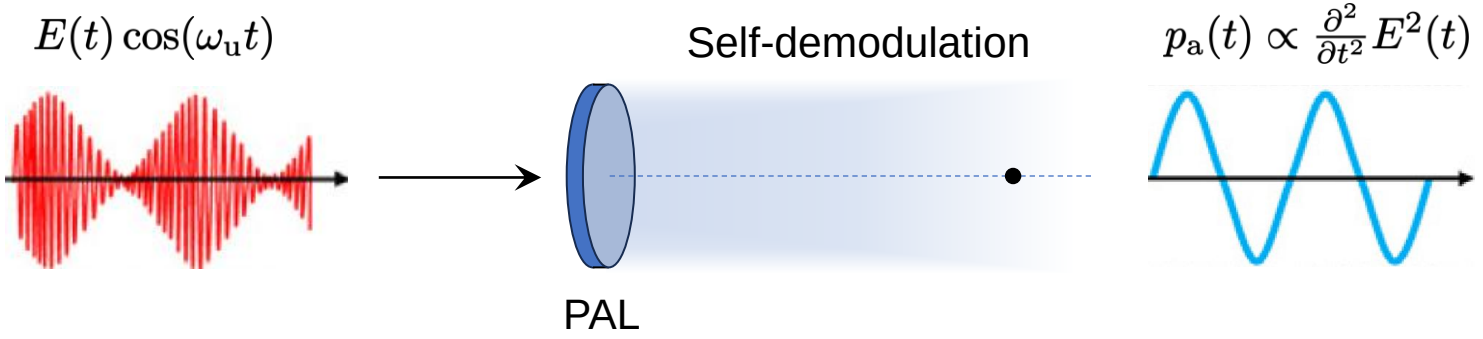
$$q(\mathbf{r}_\mathrm{v}) = \frac{\beta\omega_\mathrm{a}}{\mathrm{i}\rho_0^2 c_0^4}p_1^*(\mathbf{r}_\mathrm{v})p_2(\mathbf{r}_\mathrm{v}) \qquad p_\mathrm{a}(\mathbf{r}) = -\mathrm{i}\rho_0\omega_\mathrm{a}\iiint_V q(\mathbf{r}_\mathrm{v})g(\mathbf{r}, \mathbf{r}_\mathrm{v}; \omega_\mathrm{a})\,\mathrm{d}^3\mathbf{r}_\mathrm{v}$$

Virtual audio source

[1] J. Zhong, R. Kirby, and X. Qiu, 'The near field, Westervelt far field, and inverse-law far field of the audio sound generated by parametric array loudspeakers', The Journal of the Acoustical Society of America, vol. 149, no. 3, pp. 1524–1535, Mar. 2021.
[2] P. J. Westervelt, "Parametric acoustic array," The Journal of the Acoustical Society of America, vol. 35, no. 4, pp. 535–537, Apr. 1963.
[3] M. Červenka and M. Bednařík, "A versatile computational approach for the numerical modelling of parametric acoustic array," The Journal of the Acoustical Society of America, vol. 146, no. 4, pp. 2163–2169, Oct. 2019.
[4] J. Yang, K. Sha, W.-S. Gan, and J. Tian, "Nonlinear wave propagation for a parametric loudspeaker," IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences, vol. E87-A, no. 9, pp. 2395–2400, Sep. 2004.
[5] M. Červenka and M. Bednařík, "Non-paraxial model for a parametric acoustic array," The Journal of the Acoustical Society of America, vol. 134, no. 2, pp. 933–938, Aug. 2013.

Comparative analysis of time-delay strategies

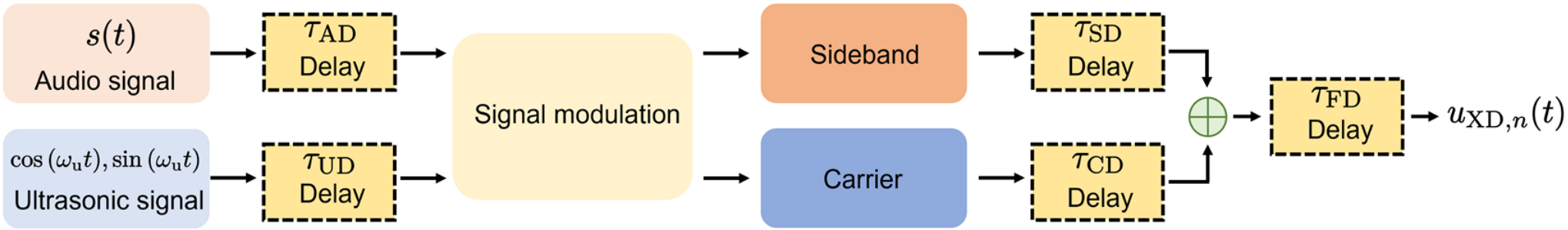
- Time-delay strategies & analysis framework
  - Berktay's far-field self-demodulation solution



  - Single sideband amplitude modulation (SSBAM)

$$u(t) = \underbrace{\cos(\omega_u t)}_{\text{Carrier}} + \underbrace{\mathcal{M}\left[\, s(t)\,\cos(\omega_u t) - \hat{s}(t)\,\sin(\omega_u t)\,\right]}_{\text{Upper sideband}}, \qquad \hat{s}(t) = \mathcal{H}\left[\, s(t)\,\right]$$

  - Time-delay strategies



[1] X. Wu, T. Zhuang, S. Li, M. Li, J.-X. Zhong, and J. Lu, 'A comparative analysis of time-delay strategies for phased-array parametric loudspeakers', IEEE Trans. Audio, Speech Lang. Process., vol. 34, pp. 3743–3753, 2026.

Comparative analysis of time-delay strategies

- Simulation framework: k-space modeling
  - FFT-accelerated k-space computation

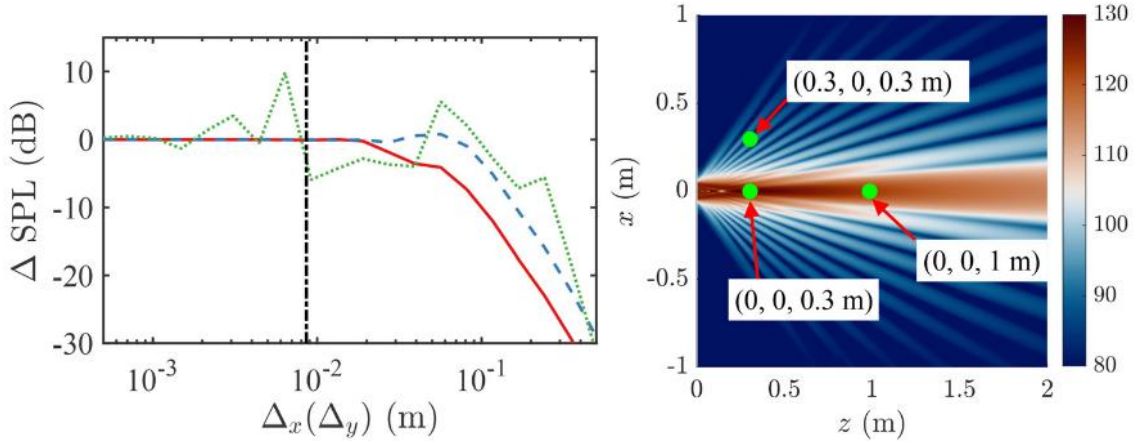
Ultrasound: $p_i(\mathbf{r}) \propto v_{i,z}(x,y) *_x *_y g_i(\mathbf{r}), \qquad i = 1,2$

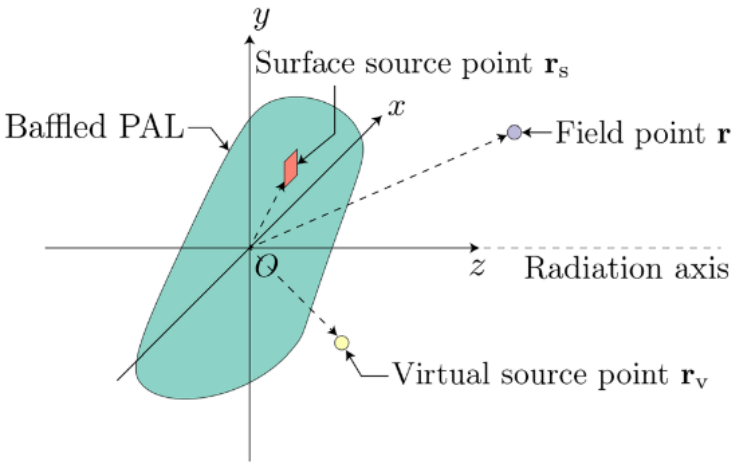
Audio: $p_{\mathrm{a}}(\mathbf{r}) \propto (p_1^* p_2) *_x *_y *_z g_{\mathrm{a}}(\mathbf{r})$

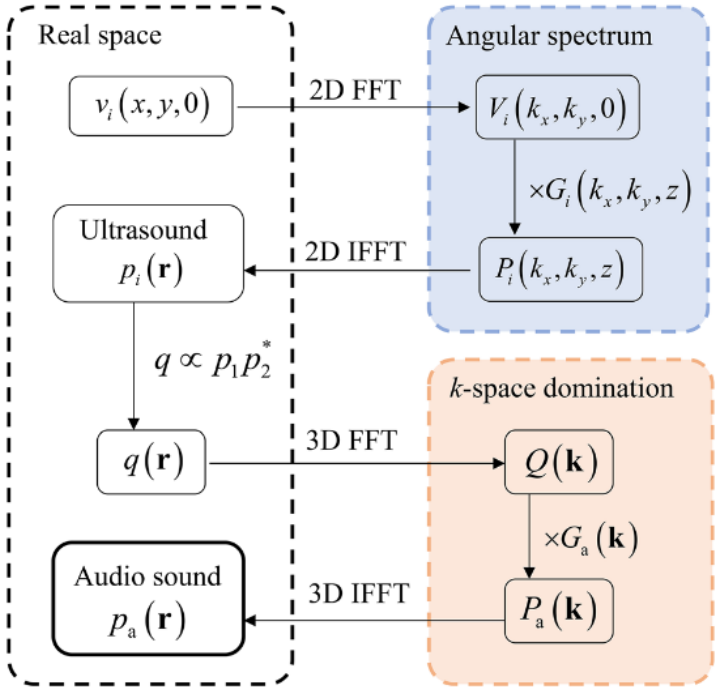
FFT-accelerated convolution reduces computational complexity.

Method	Total space complexity	Time complexity
DIM (volume integral)	$\mathcal{O}(N_x N_y N_z)$	$\mathcal{O}(N_x^2 N_y^2 N_z^2)$
k-space (FFT-based)	$\mathcal{O}(N_x N_y N_z)$	$\mathcal{O}(N_x N_y N_z \log(N_x N_y N_z))$

  - Convergence of the k-space model





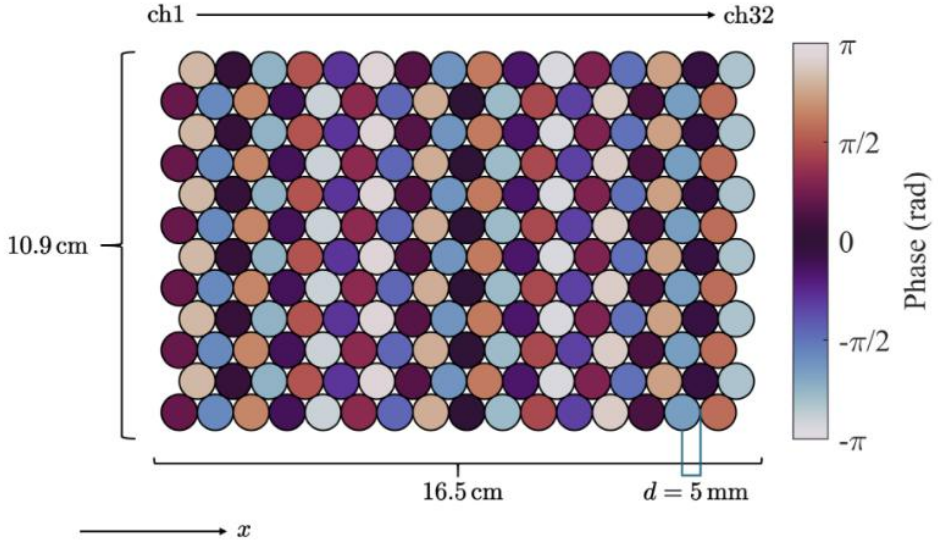


[1] T. Zhuang, L. He, F. Niu, J.-X. Zhong, and J. Lu, 'A k-space approach to modeling multi-channel parametric array loudspeaker systems', The Journal of the Acoustical Society of America, vol. 158, no. 6, pp. 4651–4661, Dec. 2025.

Comparative analysis of time-delay strategies

- Simulation setup: discrete steerable PAL array

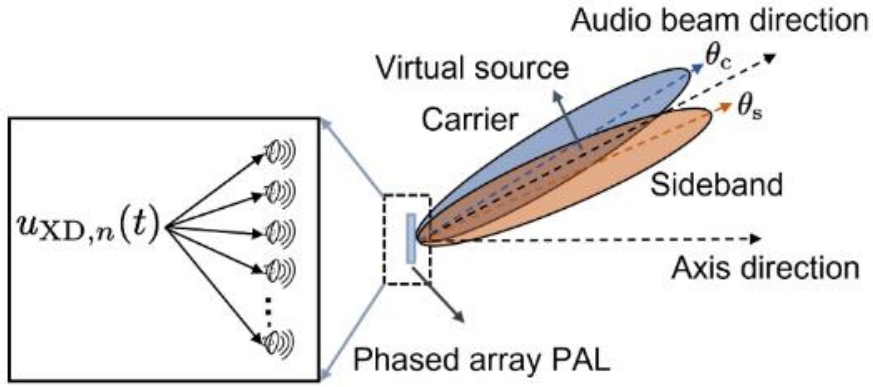
$$u(t) = \underbrace{\cos(\omega_u t)}_{\text{Carrier}} + \underbrace{\mathcal{M}\left[\, s(t)\,\cos(\omega_u t) - \hat{s}(t)\,\sin(\omega_u t)\,\right]}_{\text{Upper sideband}}, \qquad \hat{s}(t) = \mathcal{H}\left[\, s(t)\,\right]$$



Parameter	Value
Propagation medium	Air
Temperature	20 °C
Relative humidity	70%
Sound speed c_0	343 m/s
Air density ρ_0	1.21 kg/m^3
Nonlinearity coefficient β	1.2
Normal surface velocity amplitude v_0	0.121 m/s

$$\tau_n = \frac{nd}{c_0}\sin\theta_0, \quad \phi_n(\omega) = -\omega\tau_n = -\frac{\omega n d}{c_0}\sin\theta_0$$

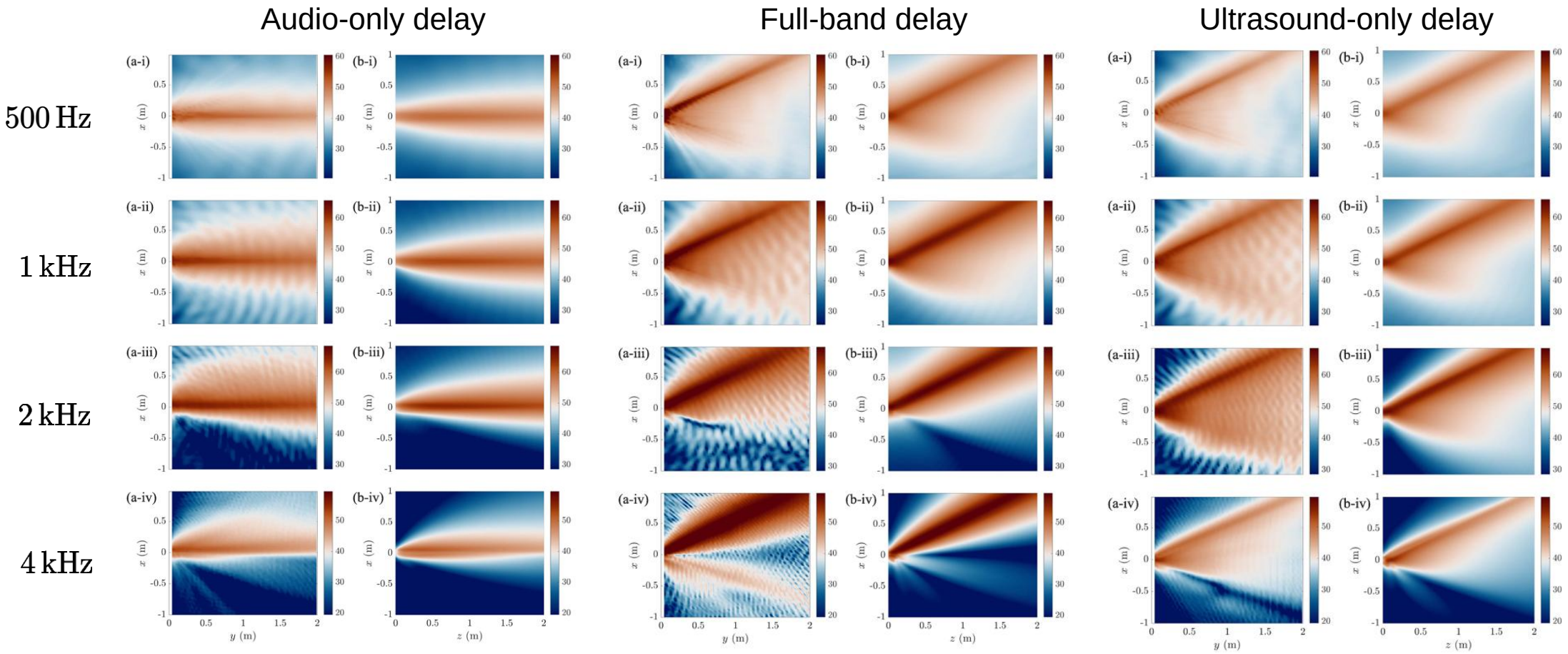
Strategy	$\phi_{1,n}$	$\phi_{2,n}$
AD	0	$-\omega_a\tau_n$
UD	$-\omega_u\tau_n$	$-\omega_u\tau_n$
CD	$-\omega_u\tau_n$	0
SD	0	$-(\omega_u+\omega_a)\tau_n$
FD	$-\omega_u\tau_n$	$-(\omega_u+\omega_a)\tau_n$



[1] X. Wu, T. Zhuang, S. Li, M. Li, J.-X. Zhong, and J. Lu, ‘A comparative analysis of time-delay strategies for phased-array parametric loudspeakers’, IEEE Trans. Audio, Speech Lang. Process., vol. 34, pp. 3743–3753, 2026.

Comparative analysis of time-delay strategies

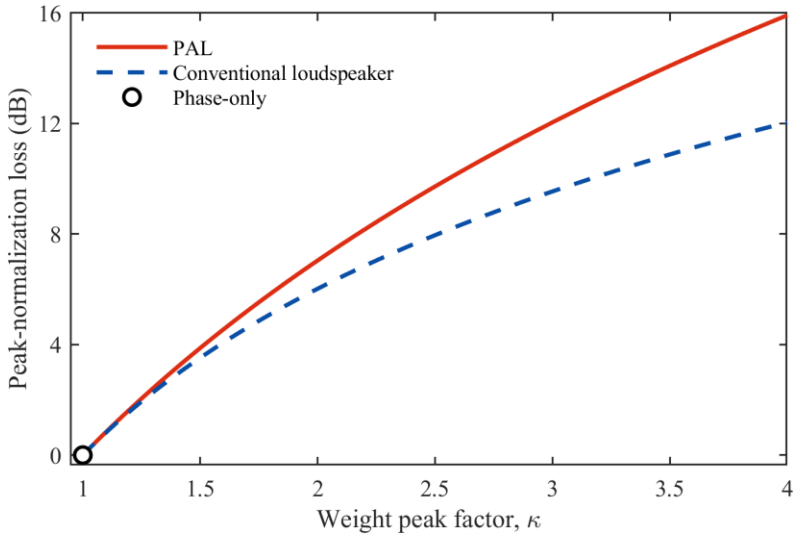
- Experimental validation and comparison
  - Measurements validate the k-space predictions (left column: exp. ; right column: sim.)



UD achieves near-FD steering performance at low implementation cost, whereas AD is ineffective.

Phase-only optimization under transmitter constraint

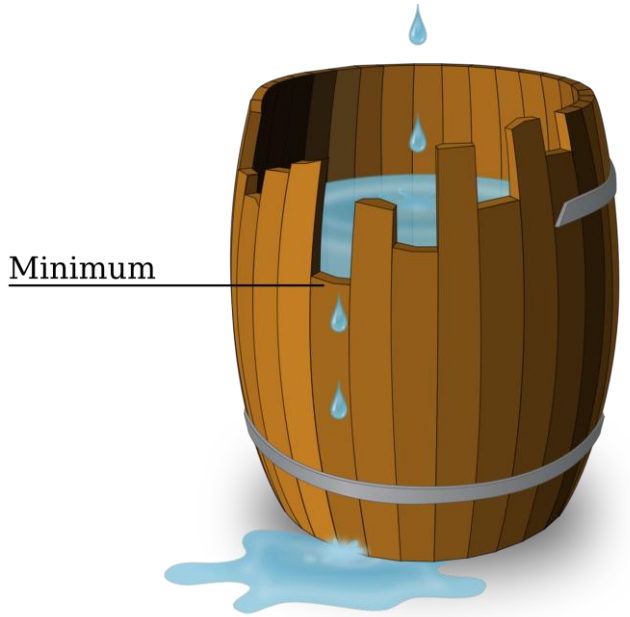
- Audible sound energy & hardware constraints
  - Low demodulation efficiency in PAL (e.g., ultrasonic-to-audio power conversion below 0.05% in air under the studied conditions)
  - Transmitter constraint & output loss
    - The channel with the largest amplitude limits the overall drive level
    - Sound field control introduces a trade-off between weight variation (control performance) and audible output energy.



$$\frac{1}{N}\sum_{n=1}^{N}|w_n|^2 = 1, \qquad \kappa = \max_{1\le n\le N}|w_n|$$

$$L_{\mathrm{L}} = 20\log_{10}\kappa \ \mathrm{dB}$$

$$L_{\mathrm{PAL}} \approx 40\log_{10}\frac{1+\kappa}{2} \ \mathrm{dB}$$



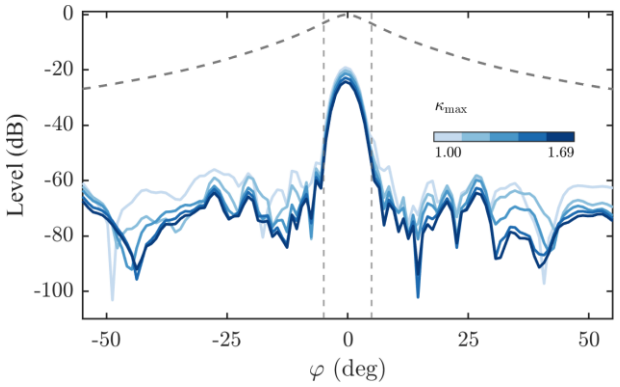
[1] M. Li, J. Zhong, Y. Jing, and J. Lu, 'A note on the audio sound power generated by a parametric array loudspeaker', The Journal of the Acoustical Society of America, vol. 154, no. 6, pp. 3899–3905, Dec. 2023.

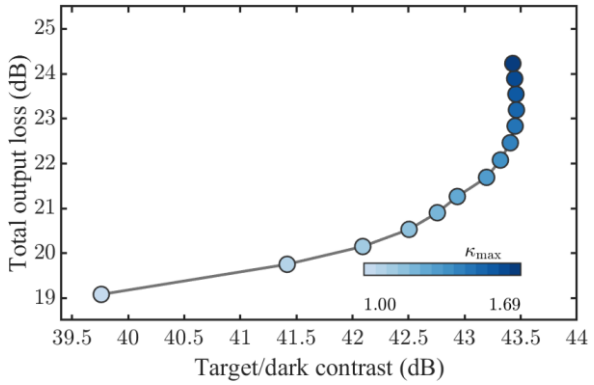
[2] Ö. Çayır and Ç. Candan, 'Transmit beamformer design with a PAPR constraint to trade-off between beampattern shape and power efficiency', Digital Signal Processing, vol. 99, p. 102674, Apr. 2020.

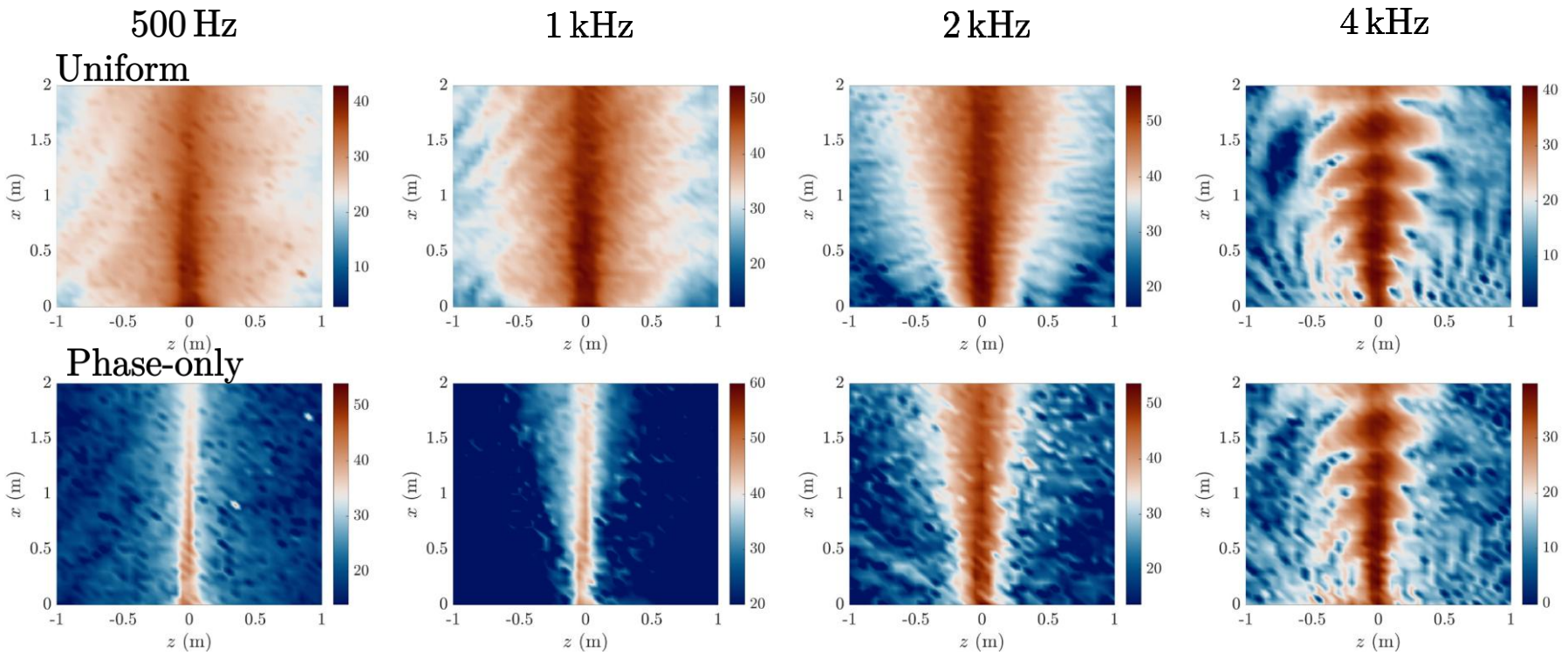
Phase-only optimization under transmitter constraint

- Control performance-output trade-off: narrow beam

$$\min_{\mathbf{w}} \; J(\mathbf{w}) \quad \text{s.t.} \quad \frac{1}{N}\|\mathbf{w}\|_2^2 = 1, \qquad \|\mathbf{w}\|_\infty \le \kappa_{\max} \qquad J_{\mathrm{MVDR}}(\mathbf{w}) = -\frac{|\mathbf{g}_0\mathbf{w}|^2}{\mathbf{w}^{\mathrm{H}}\left(\mathbf{R}_{\mathrm{D}} + \delta\mathbf{I}\right)\mathbf{w}}$$



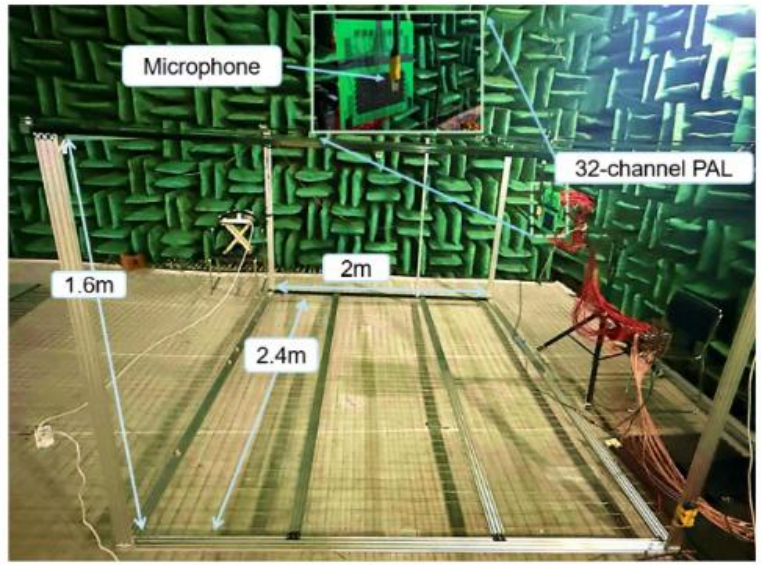




In this case, phase-only control provides higher audible output at a modest cost in control performance.

Conclusions

- Comparative analysis of time-delay strategies
  - Explained delay mechanisms and identified low-cost UD
  - Built a reusable measurement platform
  - Validated k-space modeling for PAL
- Phase-only optimization under transmitter constraint
  - Control-output trade-off
  - Increased output with a modest control-performance cost
- Future work
  - Fast PAL directivity prediction and efficient control
  - More experimental validation of phase-only control



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Thanks